Section 16.4: Green's Theorem

Vector Calculus

Warm-Up: One Problem, Three Ways

Let $\vec{F} = \langle 2xy + 1,\ x^2 \rangle$ and let C be the parabola $y = x^2$ from $(0, 0)$ to $(1, 1)$.

Evaluate $\displaystyle\int_C \vec{F} \cdot d\vec{r}$ using all three methods:

1. **Definition:** Parameterize C, compute $\vec{F}(\vec{r}(t)) \cdot \vec{r}'(t)$, integrate.
2. **Component form:** Write as $\int_C P\,dx + Q\,dy$ and substitute.
3. **FTC:** Find a potential function f with $\nabla f = \vec{F}$, then use $f(\text{end}) - f(\text{start})$.

Try all three. You should get the same answer each time. Which method is easiest?

Line Integrals Meet Double Integrals

In 16.3, conservative fields gave us $\oint_C \vec{F} \cdot d\vec{r} = 0$ for every closed curve. But what if $\vec{F}$ is **not** conservative? Green's Theorem answers this: the line integral around a closed curve equals a **double integral** over the enclosed region.

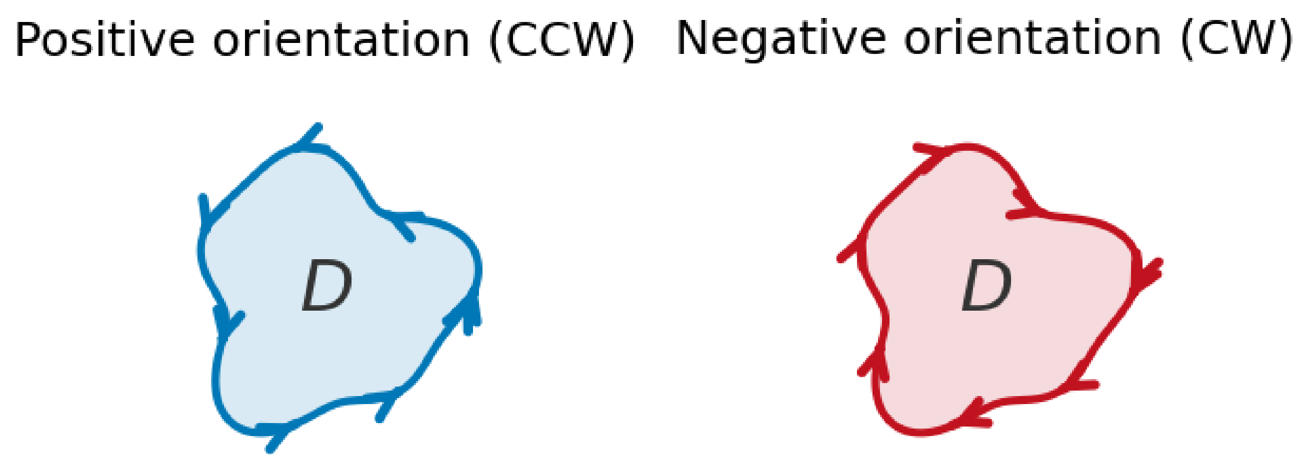
$$(\text{integral around boundary}) = (\text{integral over interior})$$

This is a higher-dimensional FTC: just as $\int_a^b f'(x)\,dx = f(b) - f(a)$ relates an interval integral to boundary values, Green's Theorem relates a region integral to a boundary integral.

Positive Orientation

Positive Orientation

A simple closed curve C is **positively oriented** if, as you traverse it, the enclosed region D stays on your **left**. In the plane: **counterclockwise**.

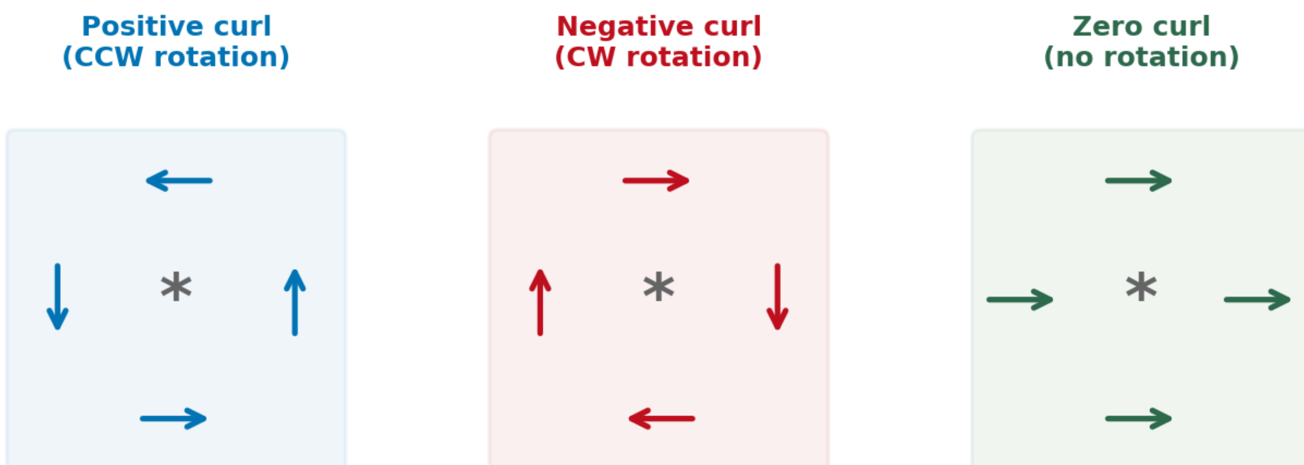


Clockwise traversal is negative orientation: $\oint_{-C} \vec{F} \cdot d\vec{r} = -\oint_C \vec{F} \cdot d\vec{r}$.

Green's Theorem requires **positive (CCW) orientation**. If the curve is CW, negate.

Micro-Circulation: What Does Local Rotation Look Like?

Imagine placing a tiny paddle wheel at a point in a vector field. Does it spin?



Look at how the arrows change across the box:

- Q **increases** left-to-right ($Q_x > 0$): the right side pushes up more than the left $\to$ CCW spin
- P **increases** bottom-to-top ($P_y > 0$): the top pushes right more than the bottom $\to$ CW spin

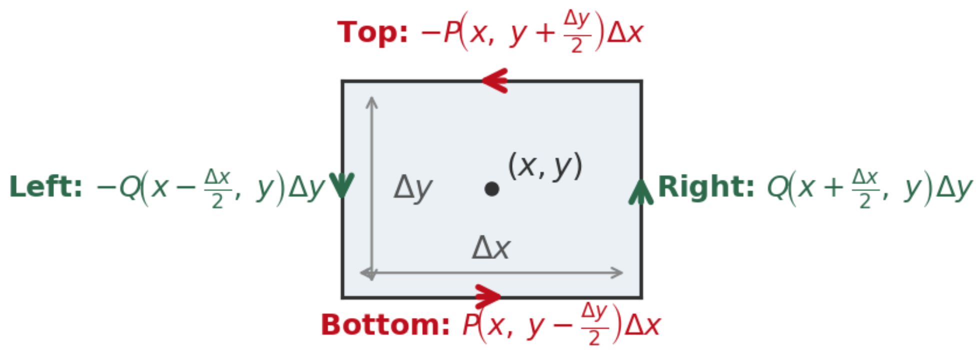
The net rotation is the competition between these: $Q_x - P_y$.

- $Q_x - P_y > 0$: net CCW rotation (positive curl)
- $Q_x - P_y < 0$: net CW rotation (negative curl)
- $Q_x - P_y = 0$: no net rotation (conservative!)

The **2D curl** $Q_x - P_y$ measures local counterclockwise rotation — the micro-circulation per unit area. It's exactly the quantity from the cross-partials test: zero curl $=$ conservative.

Making It Precise: Circulation on a Tiny Rectangle

Center a tiny rectangle at (x, y) and compute $\oint \vec{F} \cdot d\vec{r}$ around it (CCW):



Horizontal edges (top and bottom) contribute:

$$P\left(x,\, y-\frac{\Delta y}{2}\right)\Delta x - P\left(x,\, y+\frac{\Delta y}{2}\right)\Delta x \;\approx\; -\frac{\partial P}{\partial y}\,\Delta y\,\Delta x$$

Vertical edges (right and left) contribute:

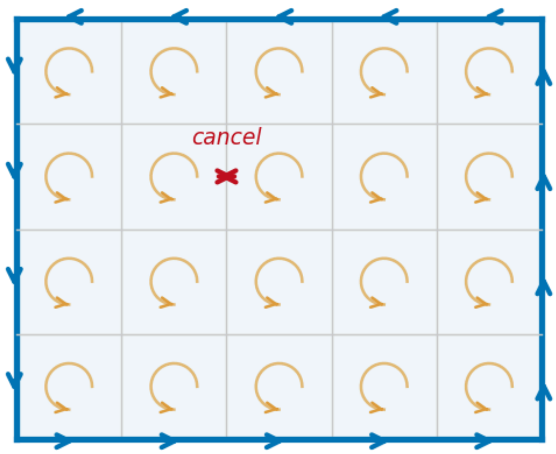
$$Q\left(x+\frac{\Delta x}{2},\, y\right)\Delta y - Q\left(x-\frac{\Delta x}{2},\, y\right)\Delta y \;\approx\; \frac{\partial Q}{\partial x}\,\Delta x\,\Delta y$$

Total circulation $\approx \left(\dfrac{\partial Q}{\partial x} - \dfrac{\partial P}{\partial y}\right)\Delta A$

The circulation per unit area around a tiny rectangle is $Q_x - P_y$. This is the **2D curl** — it measures local counterclockwise rotation.

Why Green's Theorem Works: Cancellation

Tile the whole region with tiny rectangles. Each has micro-circulation $(Q_x - P_y)\,\Delta A$.



Key insight: Adjacent cells share an edge, traversed in **opposite directions**. These cancel!

What survives? Only the **outer boundary** — edges not shared with a neighbor.

$$\sum \text{micro} \;\to\; \iint_D (Q_x - P_y)\,dA$$

$$= \oint_C P\,dx + Q\,dy$$

The interior cancels, leaving only the boundary circulation. That's Green's Theorem: macro = sum of micro.

Green's Theorem

Theorem: Green's Theorem

Let C be a positively oriented, piecewise-smooth, simple closed curve bounding a region D. If P and Q have continuous partial derivatives on an open region containing D, then

$$\boxed{\oint_C P\,dx + Q\,dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dA}$$

In words: circulation around boundary $=$ total curl over interior.

When the curl is constant (say $Q_x - P_y = k$), the integral is just $k \cdot \mathrm{Area}(D)$.

Quick Check: For $\oint_C (3y)\,dx + (5x)\,dy$: $P = 3y$, $Q = 5x$, so $Q_x - P_y = 5 - 3 = 2$. By Green's: $\oint = \iint_D 2\,dA = 2 \cdot \mathrm{Area}(D)$. When the curl is constant, the double integral is just the constant times the area!

Green's Theorem converts a line integral (hard) into a double integral (often easier), or vice versa.

Example 1: Verify Green's Theorem

Example 1. Evaluate $\oint_C xy\,dx + x^2\,dy$ where C is the rectangle $(0,0) \to (3,0) \to (3,1) \to (0,1)$ (CCW). Verify both sides.

Compute $Q_x - P_y$ and integrate over the rectangle. Then verify by computing the line integral along all four edges.

Example 2: Symmetry Shortcut

Example 2

Use Green's Theorem to evaluate $\oint_C y^4\,dx + 2xy^3\,dy$ where C is the ellipse $x^2 + 2y^2 = 2$ (CCW).

Compute $Q_x - P_y$. Before integrating, check: is the integrand odd or even on a symmetric region?

Example 3: Green's Theorem Throws Away the Hard Parts

Example 3

Evaluate $\oint_C (e^x + y^2)\,dx + (e^y + x^2)\,dy$ where C is the boundary of the region between $y = x^2$ and $y = x$ (CCW).

Compute $Q_x - P_y$. Find where $y = x^2$ and $y = x$ intersect. Set up and evaluate the double integral.

Computing Area with Green's Theorem

If we choose P and Q so that $Q_x - P_y = 1$, then $\oint P\,dx + Q\,dy = \text{Area}(D)$.
The cleanest choice: $P = -y/2$, $Q = x/2$, giving

$$\boxed{\text{Area}(D) = \frac{1}{2}\oint_C x\,dy - y\,dx}$$

This computes area from a boundary parameterization alone — no double integral.

Example 4: Area of an Ellipse

Example 4

Use the Green's Theorem area formula to find the area enclosed by the ellipse $\dfrac{x^2}{9} + \dfrac{y^2}{4} = 1.$

Parameterize the ellipse: $x = 3\cos t$, $y = 2\sin t$, $0 \le t \le 2\pi$. Compute dx and dy, then apply $A = \frac{1}{2}\oint x\,dy - y\,dx$.

Green's Theorem Explains Conservative Fields

Green's Theorem finally proves the “hard direction” of the cross-partials test from 16.3.
If $P_y = Q_x$ everywhere on a simply-connected region D, then for **any** closed curve C inside D:

$$\oint_C P\,dx + Q\,dy = \iint_D \underbrace{(Q_x - P_y)}_{=0}\,dA = 0$$

So $\vec{F}$ is path-independent, hence conservative.

Famous counterexample: $\vec{F} = \left\langle \dfrac{-y}{x^2+y^2}, \dfrac{x}{x^2+y^2} \right\rangle$ has $P_y = Q_x$ everywhere except the origin, yet $\oint \vec{F} \cdot d\vec{r} = 2\pi$ around any circle enclosing the origin. Green's Theorem doesn't apply because the hole breaks simple connectivity.

Simply connected + $P_y = Q_x \Rightarrow$ conservative. The hole in the domain is what allows $P_y = Q_x$ without conservative.

Homework

Section 16.4: 1, 3, 5, 9, 11, 15, 21